Alex Mote

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Education

The University of Manchester — Manchester, UK
MSc Robotics — *September 2026 - Present*

- Awarded the **Spärck AI Scholarship** (UK government, DSIT), a **fully funded** award for top AI talent; one of up to 10 at Manchester

BSc(Hons) Computer Science — *September 2023 - June 2026*

- **First Class Honours** with an **82% average**
- **Relevant Courses:** Machine Learning, Algorithms and Data Structures, Distributed Systems, System Architecture, IoT, Software Engineering

Bro Myrddin Welsh Comprehensive School — Carmarthen, Wales
Secondary Education — *September 2016 - June 2023*

- **A-Levels:** Maths (A*), Physics (A*), Chemistry (A*), Welsh Baccalaureate (A), **AS-Levels:** Further Maths (A)
- Awarded the ***Bro Myrddin Award*** for achieving the **highest A-Level results** in the cohort

Projects

RiskDrift | *Python, FinBERT, Streamlit* — April 2026

- Built an NLP pipeline that encodes SEC 10-K Item 1A risk disclosures with FinBERT embeddings and flags statistically significant language regime changes via per-company cosine similarity z-scores, validated against 6-month forward returns
- Implemented the FinBERT embedding module: designed a sliding-window mean-pooling strategy over 512-token chunks to encode arbitrarily long risk disclosures into single 768-dimensional document vectors, enabling cosine similarity comparison across a company's full filing history
- Reached the **national finals (top 5)** of the **CFA Institute AI Investment Challenge 2026** in a 4-person team, presenting to a panel of senior industry judges in London

GPU Sequence Alignment (Bioinformatics) | *C++, CUDA* — September 2025 – April 2026

- Engineered a parallelized protein Database Search algorithm using CUDA, achieving a **5.8x speedup** in GCUPS (Giga Cell Updates Per Second) compared to the industry standard CPU implementation (Parasail) for global alignment
- Ensured algorithmic correctness by building automated **regression testing** that checks every GPU score against Parasail on each run, alongside Catch2 unit tests
- Established a rigorous **benchmarking** framework using **CUDA events** to quantitatively evaluate optimization strategies (e.g., coalesced memory access, inter- vs. intra-sequence parallelism, hybrid kernel routing) against a naive GPU baseline

High-Performance AI Hex Agent | *C++, Python* — November – December 2025

- Collaborated to build a competitive game agent using Monte Carlo Tree Search (MCTS) with pattern-based simulations
- Optimized the C++ evaluation engine using bitboards, adjacency lookup tables, and a compact tree-node layout, resulting in a **10x increase** in simulations per second against a baseline implementation
- Developed a benchmarking framework to measure simulations per second and win-rate of new features against the current agent, enabling data-driven optimization decisions

Employment

Presiding Officer — May 2026
Carmarthenshire County Council — *Trelech, Wales*

- Managed the polling process for **400** registered voters, ensuring a fair, efficient, and professional experience for all
- Developed problem-solving skills by assisting voters with disabilities in accessing the polling station and casting votes

Tesco Colleague — June 2023 – September 2025
Tesco — *Carmarthen, Wales*

- Served **200+** customers each shift in a fast paced environment, maintaining accuracy and efficiency, while remaining friendly and assisting with customer queries
- Cooperated with team members to meet daily targets and quickly adapted to different roles when necessary

Volunteering & Leadership

Peer-Assisted Study Sessions (PASS) Leader — September 2024 – May 2025
University of Manchester Computer Science Department — *Manchester, UK*

- Cultivated effective communication by coordinating weekly meetings with **20 first-year students**, offering **mentorship** and academic support to assist with the transition into higher education
- Enhanced adaptability and interpersonal skills by providing tailored guidance based on each student's unique needs

Officer Cadet — October 2023 – June 2025
MASUAS, Royal Air Force — *RAF Woodvale, Merseyside*

- Improved **leadership** skills through **military training exercises**, including flight training, mountain rescue and first aid
- Collaborated with other members to plan and execute squadron events and activities, such as **fundraising** and **charity** events, fostering teamwork and organisational skills

Technical Skills

Languages: Java, C++, System Verilog, SQL (PostgreSQL), Python, Javascript, HTML/CSS
Tools & Frameworks: Git, Docker, AWS, Linux/Bash, GitHub Actions, Spring Boot, React, JUnit, Testcontainers, CUDA, Catch2